

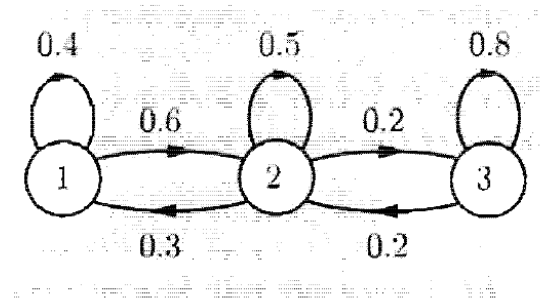
Problem 1. The times between successive customer arrivals at a facility are independent and identically distributed random variables with the following PMF:

$$p(k) = \begin{cases} 0.2, & k=1, \\ 0.3, & k=3, \\ 0.5, & k=4, \\ 0, & \text{otherwise.} \end{cases}$$

Construct a four-state Markov chain model that describes the arrival process. In this model, one of the states should correspond to the times when an arrival occurs.

Problem 2 . A professor gives tests that are hard, medium, or easy. If she gives a hard test, her next test will be either medium or easy, with equal probability. However, if she gives a medium or easy test, there is a 0.5 probability that her next test will be of the same difficulty, and a 0.25 probability for each of the other two levels of difficulty. Construct an appropriate Markov chain and find the steady-state probabilities.

Problem 3. Consider the Markov chain in Fig. 7.22. Let us refer to a transition that results in a state with a higher (respectively, lower) index as a birth (respectively, death). Calculate the following quantities, assuming that when we start observing the chain, it is already in steady-state.



- For each state i , the probability that the current state is i .
- The probability that the first transition we observe is a birth.
- The probability that the first change of state we observe is a birth.
- The conditional probability that the process was in state 2 before the first transition that we observe, given that this transition was a birth.
- The conditional probability that the process was in state 2 before the first change of state that we observe, given that this change of state was a birth.
- The conditional probability that the first observed transition is a birth given that it resulted in a change of state.
- The conditional probability that the first observed transition leads to state 2, given that it resulted in a change of state.

Problem 4. There are m classes offered by a particular department, and each year, the students rank each class from 1 to m , in order of difficulty, with rank m being the highest. Unfortunately, the ranking is completely arbitrary. In fact, any given class is equally likely to receive any given rank on a given year (two classes may not receive the same rank). A certain professor chooses to remember only the highest ranking his class has ever gotten.

- Find the transition probabilities of the Markov chain that models the ranking that the professor remembers.
- Find the recurrent and the transient states.
- Find the expected number of years for the professor to achieve the highest ranking given that in the first year he achieved the i th ranking.

Problem 5. Consider the Markov chain specified in Fig. 7.25.

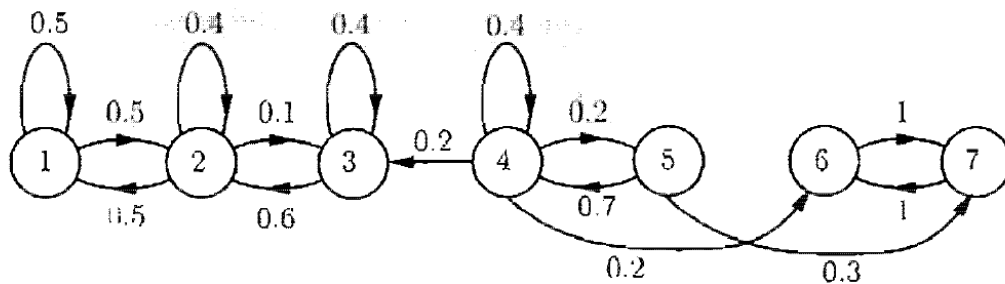


Figure 7.25: Transition probability graph for Problem 30.

- Identify the transient and recurrent states. Also, determine the recurrent classes and indicate which ones, if any are periodic.
- Do there exist steady-state probabilities given that the process starts in state 1? If so, what are they?
- Do there exist steady-state probabilities given that the process starts in state 6? If so, what are they?
- Assume that the process starts in state 1 but we begin observing it after it reaches steady-state.
 - Find the probability that the state increases by one during the first transition we observe.
 - Find the conditional probability that the process was in state 2 when we started observing it, given that the state increased by one during the first transition that we observed.
 - Find the probability that the state increased by one during the first change of state that we observed.
- Assume that the process starts in state 4.
 - For each recurrent class, determine the probability that we eventually reach that class.
 - What is the expected number of transitions up to and including the transition at which we reach a recurrent state for the first time?